Surds (1)

Worksheet Exercise 1

1. $2\sqrt{8} - 4\sqrt{32} + 3\sqrt{50}$

$= 2\sqrt{4\times2} - 4\sqrt{16\times2} + 3\sqrt{25\times2}$

$= (2)(2)(\sqrt{2}) - (4)(4)(\sqrt{2}) + (3)(5)(\sqrt{2})$

$= 4\sqrt{2} - 16\sqrt{2} + 15\sqrt{2}$

$= 4\sqrt{2}$

2. $\sqrt{50}\,(\sqrt{18} + \sqrt{32})$

$= \sqrt{25\times2}(\sqrt{9\times2} + \sqrt{16\times2})$

$= (5)(\sqrt{2})[3\sqrt{2} + 4\sqrt{2}]$

$= (5\sqrt{2})(7\sqrt{2})$

$= 70$

3. $(2-\sqrt{3})^2 + (2+\sqrt{3})^2$

$= (2-\sqrt{3})(2-\sqrt{3}) + (2+\sqrt{3})(2+\sqrt{3})$

$= [4 - 4\sqrt{3} + 3] + [4 + 4\sqrt{3} + 3]$

$= 7 + 7 = 14$

4. $\left(\frac{\sqrt{8}}{2} - \sqrt{18} + \sqrt{72}\right)\left(\frac{2}{\sqrt{2}} + \frac{4}{\sqrt{8}}\right)^{-1}$

$= \left(\frac{2\sqrt{2}}{2} - \sqrt{9\times2} + \sqrt{9\times8}\right)\left(\frac{\sqrt{2}\sqrt{2}}{\sqrt{2}} + \frac{4}{2\sqrt{2}}\right)^{-1}$

$= \left(\sqrt{2} - 3\sqrt{2} + 3(\sqrt{8})\right)\left(\sqrt{2} + \frac{\sqrt{2}\sqrt{2}}{\sqrt{2}}\right)^{-1}$

$= (\sqrt{2} - 3\sqrt{2} + 6\sqrt{2})(2\sqrt{2})^{-1}$

$= \frac{4\sqrt{2}}{2\sqrt{2}} = 2$

Exercise 2

1. $162 - 2\sqrt[3]{x^4} = 0$

$-2x^{4/3} = -162$

$2x^{4/3} = 162$

$x^{4/3} = 81 = 9^2 = 3^4$

$x = (3^4)^{3/4} = 27$

2. $3^{2x} - 3^{x+2} + 9 = 3^x$

$3^{2x} - (3^x)(3)^2 + 3^2 - 3^x = 0$

$3^x(3^x - 3^2) + 3^2 - 3^x = 0$

$3^x(3^x - 3^2) - (3^x - 3^2) = 0$

$(3^x - 1)(3^x - 3^2) = 0$

$3^x - 1 = 0$ or $3^x - 3^2 = 0$

$3^x = 1 = 3^0$ or $3^x = 3^2$

$x = 0$ or $x = 2$

Exponents

1. $81^{1/4} \times 27^{-2/3}$

$= (3^4)^{1/4} \times (3^3)^{-2/3}$

$= 3 \times 3^{-2} = \frac{3}{9} = \frac{1}{3}$

2. $\frac{6^{6x} \times 9^{3x}}{(54^{4x})(\frac{1}{4})^{2-x}}$

$54 = 6\times9$
$= 2\times3\times3^2$
$= 2\times3^3$
$4 = 2^2$

$= \frac{2^{6x}\times3^{6x}\times3^{6x}}{2^{4x}\times3^{12x}\times(2^{-2})^{2-x}}$

$= \frac{2^{6x}\times3^{12x}}{2^{4x}\times3^{12x}\times2^{2x-4}}$

$= 2^{6x-4x-2x+4}\times3^{12x-12x}$

$= 2^4 = 16$

3. $\frac{2^{n+1}}{(2n)^{n-1}} \div \frac{4^{n+1}}{(2^{n-1})^{n+1}}$

$= \frac{2^{n+1}}{2^{n-1}n^{n-1}} \times \frac{2^{(n-1)(n+1)}}{2^{2n+2}}$

$= \frac{2^{n+1+n^2-1-(n-1)-(2n+2)}}{n^{n-1}}$

$= \frac{2^{n+1+n^2-1-n+1-2n-2}}{n^{n-1}}$

$= \frac{2^{n^2-2n-1}}{n^{n-1}}$

$= 2^{n^2-2n-1}\times n^{1-n}$

4. $$\frac{18^x(2.3^{1-x})}{2^{x-1}}$$
$$= 2^x \times 3^{2x} \times 2 \times 3^{1-x} \times 2^{1-x}$$
$$= 2^{x+1+1-x} \times 3^{2x+1-x}$$
$$= 2^2 \times 3^{x+1}$$
$$= 4 \times 3 \times 3^x$$
$$= 12 \times 3^x$$

5. $$125^{2/3} + 16^{3/4} - 8^{-2/3}$$
$$= (5^3)^{2/3} + (2^4)^{3/4} - (2^3)^{-2/3}$$
$$= 5^2 + 2^3 - 2^{-2}$$
$$= 25 + 8 - \frac{1}{4}$$
$$= 33 - \frac{1}{4} = 32\frac{3}{4}$$

6. $$\frac{9^{\frac{1}{2}x} - 3^{x-1}}{3^{x+2}} = \frac{3^x - (3^x)3^{-1}}{3^x 3^2}$$
$$= \frac{3^x(1-\frac{1}{3})}{(3^x)(3^2)} = \frac{2}{3} \times \frac{1}{9}$$
$$= \frac{2}{27}$$

7. $$\frac{3.2^m - 4.2^{m+2}}{2^m - 2^{m+1}}$$
$$= \frac{2^m(3 - 4.2^2)}{2^m(1-2)}$$
$$= \frac{3-16}{-1} = 13$$

8. $$\frac{2^{2x} - 2^x - 6}{2^x + 2}$$
$$= \frac{(2^x+2)(2^x-3)}{2^x+2}$$
$$= 2^x - 3$$

Exercise 2

1. $$4.8^{x-1} = \frac{2}{16^x}$$
$$4.(2^3)^{x-1} = \frac{2}{2^{4x}}$$
$$2^2.2^{3x-3} = 2.2^{-4x} = 2^{1-4x}$$
$$2^{3x-3+2} = 2^{1-4x}$$
$$2^{3x-1} = 2^{1-4x}$$
$$3x - 1 = 1 - 4x$$
$$7x = 2$$
$$x = \frac{2}{7}$$

2. $$3^{x+1} - 3^{x-1} = 24\sqrt{3}$$
$$3^x.3 - 3^x.3^{-1} = 24 \times 3^{\frac{1}{2}}$$
$(\times 3)$ $$3^x.3^2 - 3^x = 8 \times 3^{5/2}$$
$$3^x(3^2 - 1) = 8 \times 3^{5/2}$$
$$3^x = 3^{5/2}$$
$$x = \frac{5}{2}$$

3. $$2^{2x+1} = -4 + 9.2^x$$
$$2^{2x}.2 - 9.2^x + 4 = 0$$
$$(2^x - 4)(2.2^x - 1) = 0$$
$2^x = 4$ or $2.2^x = 1$
$2^x = 2^2$ or $2^{x+1} = 2^0$
$x = 2$ or $x = -1$

4. $$5^x - 24 = 5^{2-x}$$
$$5^x - \frac{5^2}{5^x} - 24 = 0$$
$(\times 5^x)$ $$5^{2x} - 24.5^x - 25 = 0$$
$$(5^x + 1)(5^x - 25) = 0$$
$5^x = -1$ or $5^x = 5^2$

Not possible. $x = 2$

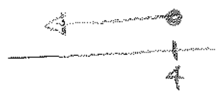
Leave Ex 2
number 5.

Inequalities

1. $-3(r-4) \geq 0$

$(r-4) \leq 0$

$r \leq 4$

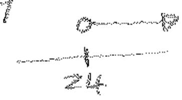


2. $\frac{-9+a}{15} > 1$

$-9+a > 15$

$a > 15+9$

$a > 24$

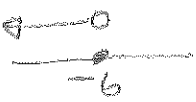


3. $-1 > \frac{12+x}{4}$

$-4 > 12+x$

$-4-12 > x$

$\therefore\ x < -16$



4. $6-4(6n+7) \geq 122$

$6-24n+28 \geq 122$

$-24n \geq 122-28-6$

$-24n \geq 88$

$n \leq \frac{88}{24} = \frac{11}{3}$

5. $-x \geq -x + 7(x-2)$

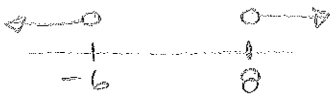
$-x \geq -x + 7x - 14$

$7x \leq 14$

$x \leq 2$

6. $m-2 < -8$ or $\frac{m}{8} > 1$

$\Rightarrow m < -6$ or $m > 8$

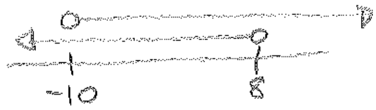


$\Rightarrow m \in (-\infty,-6) \cup (8,\infty)$

7. $-1 < 9+n < 17$

$\Rightarrow -1 < 9+n$ and $9+n < 17$

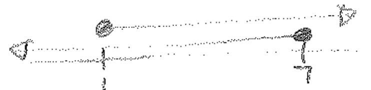
$\Rightarrow n > -10$ and $n < 8$



$\Rightarrow m \in (-10, 8)$

8. $x+8 \geq 9$ and $\frac{x}{7} \leq 1$

$x \geq 1$ and $x \leq 7$



$\Rightarrow x \in [1,7]$

9. $2n+7 \geq 27$ or $3+3n \leq 30$

$2n \geq 20$ or $3n \leq 27$

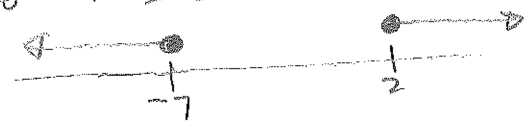
$n \geq 10$ $n \leq 9$

$\Rightarrow x \in (-\infty, 9] \cup [10,\infty)$

10. $8r-5 \geq 6r-1$ or $4+4r \leq 3r-3$

$\Rightarrow 8r-6r \geq 4$ or $r \leq -7$

$\Rightarrow r \geq 2$ or $r \leq -7$



$\Rightarrow r \in (-\infty,-7] \cup [2,\infty)$